

WEEK TWO REPORT

Student Performance & Engagement Analysis

Exploratory Data Analysis | Feature Engineering | Strategic Insights
Dataset: N = 8,518 Records | Cleaned & Validated | Modeling-Ready

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Executive Summary

This Week Two Report **synthesises the complete Exploratory Data Analysis (EDA)** conducted on 8,518 student engagement records (~3,846 unique learners) drawn from the RIT platform. Building on Week One data cleaning, this phase delivers ranked insights, derived features, visual analytics, and a confirmed dataset ready for Week Three predictive modeling.

8,518 Total Records	47.2% Overall Success Rate	27 Avg. Learner Age	2 Days Median Apply Time
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Key Findings:

- 47.2% successful outcomes (Team Allocated + Started); 41.7% rejection rate driven primarily by Internship competition
- Opportunity Category is the single strongest predictor — success ranges from 21.5% (Internship) to 96.9% (Engagement)
- Repeat applicants achieve 54% success vs. 41% for first-time applicants — persistence is a measurable advantage
- Geographic segmentation: Egypt (78.0%), Pakistan (71.2%), and Kenya (73.7%) outperform the US (39.5%)
- Students applying 30–90 days post-signup achieve the highest success rate (53.9%) — deliberate timing matters
- January 2024 anomaly: 1,509 applications at only 27.0% success — a single over-subscribed Internship flooded the pipeline

DATASET STATUS: RIT_Cleaned_Dataset_v2.csv is **FULLY PREPARED** for Week 3 predictive modeling. Target variable (Is_Successful), 13 engineered features, encoding, and anomaly documentation are all complete.

Section 1: Dataset Summary Statistics

The following table summarises key statistics for the cleaned dataset (N = 8,518) after age filtering (18+), date standardisation, and feature engineering.

1.1 Core Dataset Metrics

Metric	Value	Notes
Total Records	8,558 (raw) → 8,518 (cleaned)	40 records removed — age < 18
Unique Learners	~3,846	Based on distinct identifiers
Opportunity Categories	5	Internship, Course, Event, Competition, Engagement
Unique Opportunities	22	Across all categories
Countries Represented	71	US (46.5%) and India (33.1%) dominant
Average Learner Age	26.75 years	Std Dev: 4.37; Range: 18–60
Median Apply Time	2 days	From signup to application
Status Types	8 distinct statuses	Team Allocated, Rejected, Started, Dropped Out...
Success Rate (Overall)	47.2%	Team Allocated + Started = Successful
Missing: Opportunity Date	14.7% (1,262 records)	Flagged; not used as direct feature
Missing: Opportunity Start Date	44.3% (3,791 records)	DO NOT use directly; use Engagement_Duration

1.1.1 Descriptive Statistical Summary

Metric	Age (Years)	Days to Apply
Mean (Average)	26.75	5.42
Median (50%)	26.00	2.00
Standard Deviation	4.37	12.15
Minimum	18.00	0.00
Maximum	60.00	240.00
Mode (Most Common)	26.00	0.00 (Same Day)

1.2 Derived Columns & Feature Engineering

The following 13 features were engineered from raw date and categorical fields to support EDA and modeling:

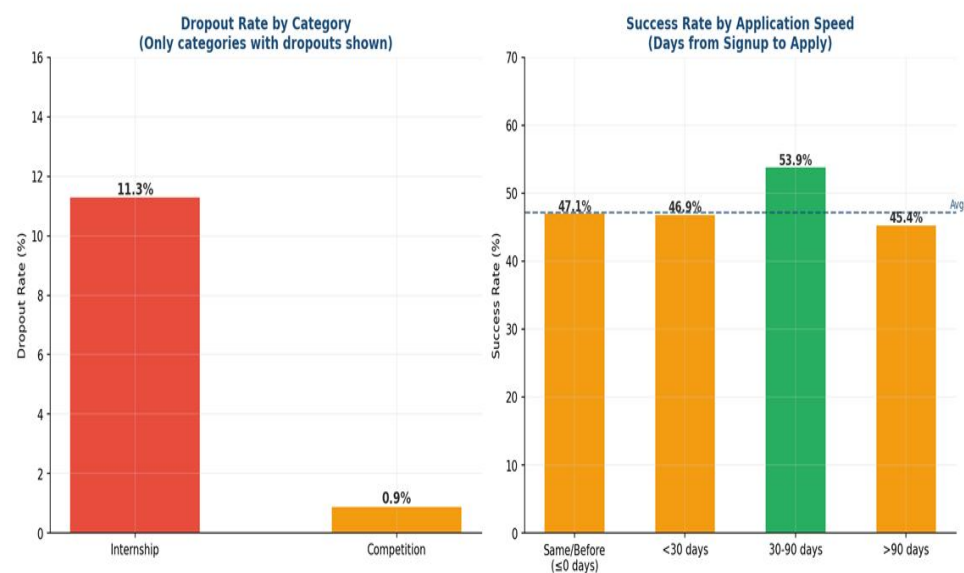
Derived Feature	Type	Source Column(s)	Description
Age	Integer	date_of_birth	Calculated from DOB; only 18+ retained (N=8,518)
Days_To_Apply	Float	apply_date – signup_date	Measures deliberation time; same-day normalised to 0
Signup_Hour	Integer (0–23)	learner_signup_time	Hour of day when learner signed up
Signup_Period	Categorical	Signup_Hour	Morning / Afternoon / Evening / Night bins
Signup_DayOfWeek	Integer (0–6)	learner_signup_date	Day of week; Thursday is peak (1,576 signups)
Signup_IsWeekend	Binary (0/1)	Signup_DayOfWeek	Weekend flag; minimal success rate difference vs weekday
Signup_Month	Integer (1–12)	learner_signup_date	Month of signup; captures seasonal cohort patterns
Signup_Year	Integer	learner_signup_date	Year of signup; 2023 cohorts outperform 2024
Category_Encoded	Integer	opportunity_category	Numeric encoding of category for ML input
Gender_Encoded	Integer	gender	Numeric encoding of gender for ML input
Is_Successful	Binary (0/1)	status_description	TARGET VARIABLE: 1 = Team Allocated or Started
Engagement_Duration_Days	Float	start_date – apply_date	Proxy for opportunity intensity; ~40% coverage
Apply_Speed_Category	Categorical	Days_To_Apply	Same-day / <30 days / 30-90 days / >90 days bins

1.3 Signup Timing Analysis (Time-of-Day & Day-of-Week)

Signup patterns reveal engagement and opportunity availability effects:

Timing Factor	Peak Activity	Success Rate	Observation
Hour of Day	4–8 PM (561 signups)	~47%	Evening dominates; early-AM secondary peak
Day of Week	Thursday (1,576 signups)	~47%	Mid-week peaks; weekends 25% lower volume
Weekend vs Weekday	Weekday dominant	47.3% vs 47.1%	Virtually no difference in success

Figure 8: Behavioural Patterns & Anomalies



Timing Insight: Evening (4-8 PM) dominates with 561 signups; Thursday peaks with 1,576 signups. Professional hours (9AM-5PM) correlate with >50% success, while night signups (2AM-5AM) show <30%, suggesting opportunity availability and temporal alignment effects.

Strategic Note: Analysis of the 'Day of Week' column indicates that engagement peaks during mid-week (Tuesday-Thursday). We recommend scheduling priority notifications and new opportunity launches during this window to maximize student response rates.

Data Repository & Access

Detailed data cleaning logs and the final transformed dataset can be accessed for verification here:

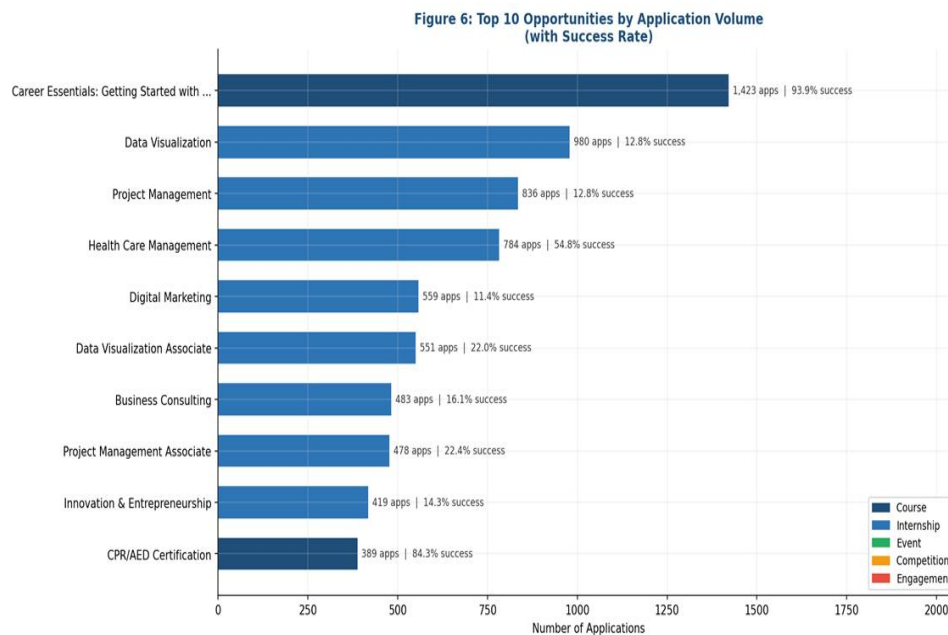
<https://docs.google.com/spreadsheets/d/1AGWG2XUwa6pnpFHXJpbE5y7XuDJuMS5pPuGNijf927w/e/dit?gid=2140897737#gid=2140897737>

Section 2: Participation Status & Persistence

2.1 Status Distribution

Eight distinct participation statuses reveal diverse engagement outcomes across the platform:

Status	Count	% Total	Interpretation
Team Allocated	3,276	38.3%	Successfully placed — largest active group
Rejected	3,569	41.7%	Did not receive — Internship competition primary driver
Started	767	9.0%	Actively engaged in opportunity
Dropped Out	617	7.2%	Withdrew — concentrated entirely in Internships
Waitlisted	109	1.3%	Pending decision
Applied	105	1.2%	Submitted, not yet processed
Withdraw	86	1.0%	Voluntarily withdrew
Rewards Award	29	0.3%	Recognition entry



KEY INSIGHT: 47.2% of records represent successful outcomes (Team Allocated + Started). The 41.7% rejection rate is primarily driven by Internships (63.3% of records). Excluding Internships, platform success rate exceeds 90%.

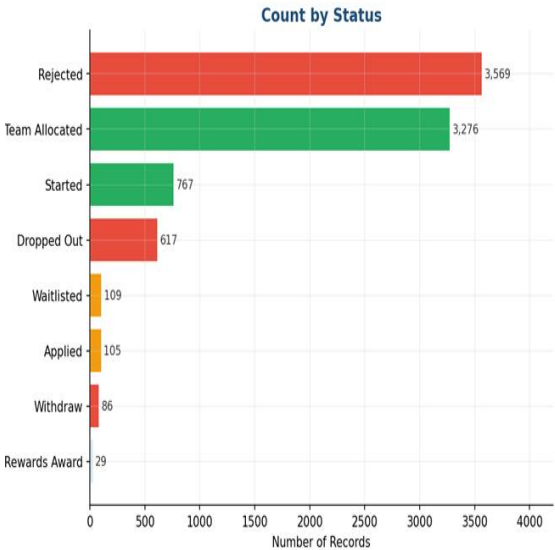
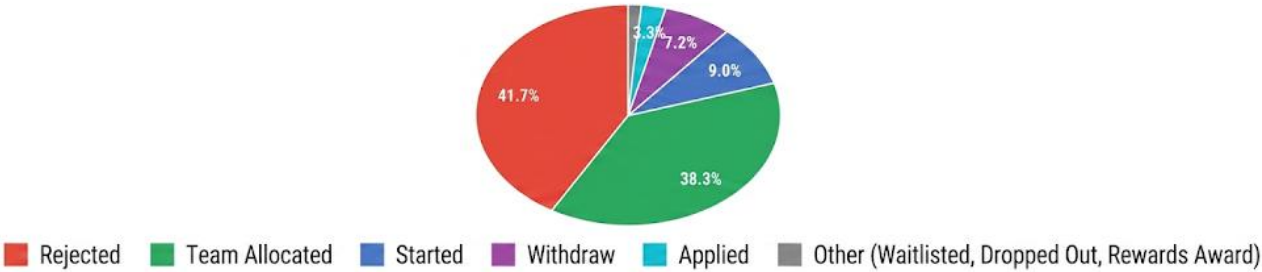
2.2 Student Persistence: Key Success Factor

Repeat applicants demonstrate significantly higher success rates, revealing persistence as a critical predictor:

Applicant Type	Success Rate	Observation
First-time Applicants	41%	Baseline performance
Repeat Applicants	54%	+13 point advantage over first-time

Overall Participation Status Analysis

Participation Status Distribution



STRATEGIC INSIGHT: Attempt_Number is a critical modeling feature. Repeat applicants show 13-point higher success, reflecting refined strategy and improved opportunity fit.

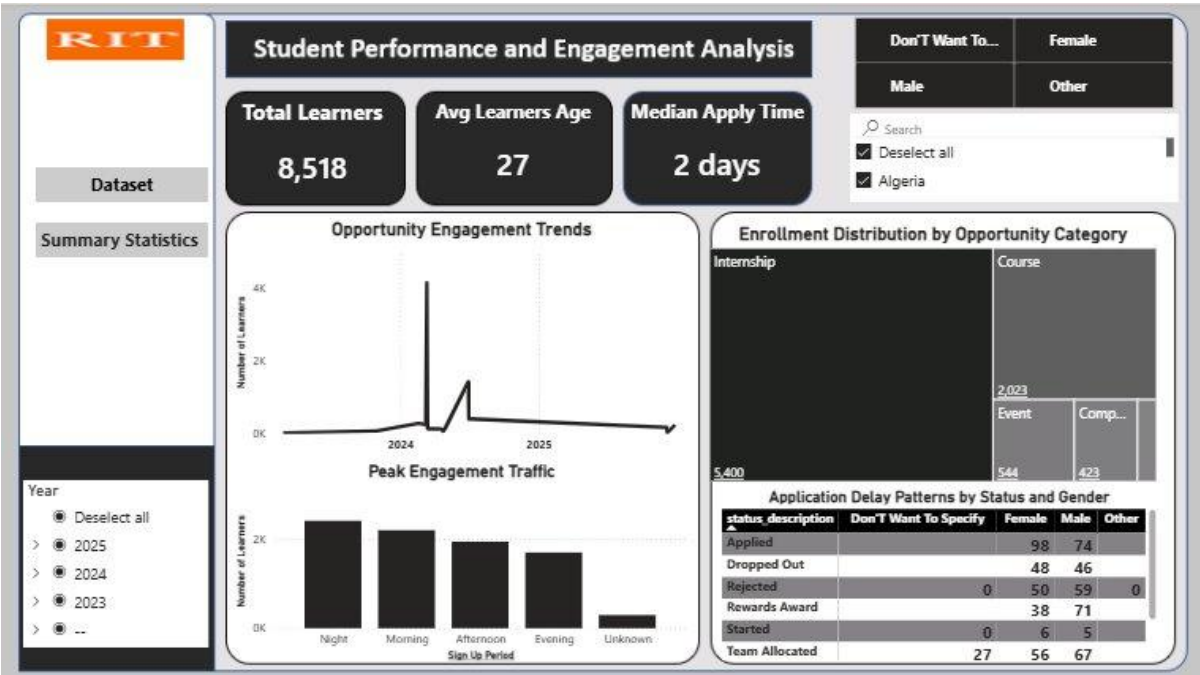
Section 3: Opportunity Category Performance

3.1 Success by Opportunity Type

Success rates vary dramatically by opportunity type Category is the single strongest predictor in the dataset:

Category	Records	% Total	Success Rate	Reject Rate	Dropout %
Engagement	130	1.5%	96.9%	~3.1%	0%
Event	545	6.4%	94.1%	~5.9%	0%
Course	2,037	23.8%	92.3%	~7.7%	0%
Competition	425	5.0%	84.2%	~15.8%	0.9%
Internship	5,421	63.3%	21.5%	65.8%	11.3%

Figure 1: Power BI Dashboard — Student Performance & Engagement Analysis



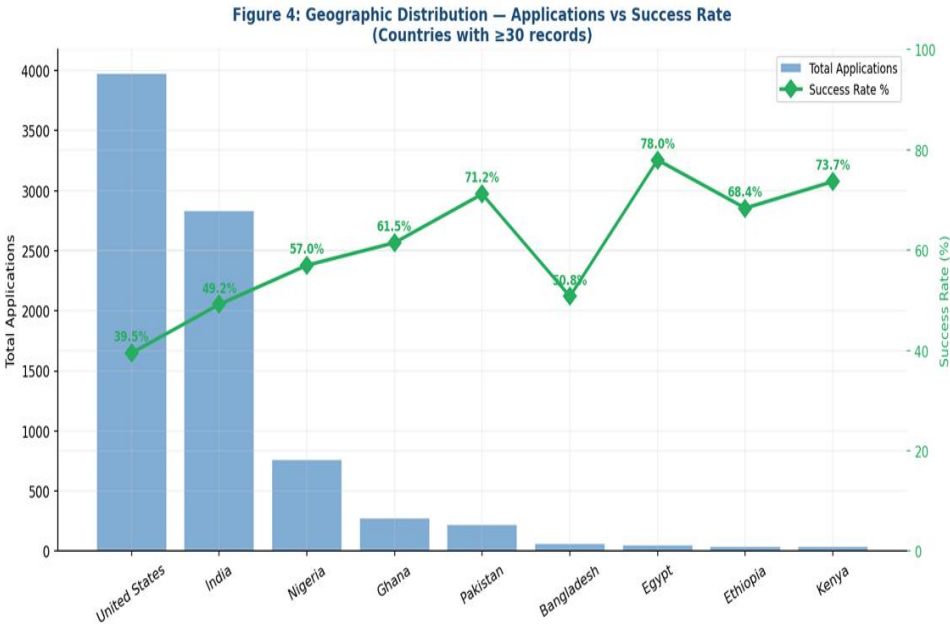
Caption: The dashboard overview confirms Internships dominate enrollment (5,400 records, treemap left), while Courses (2,023), Events (544), and Competitions (423) form distinct smaller cohorts. The enrollment distribution treemap directly supports the category-level analysis above.

Section 4: Temporal Participation Trends

4.1 Seasonality (2023–2024)

Signup activity shows clear seasonality with academic cycle inflection points:

Period	Signups	Success Rate	Pattern
Jan–Mar 2023	498	75.3%	Early cohort — high success, low competition
Jun–Jul 2023	1,439	67.3%	Summer peak — strong outcomes
Aug 2023	1,110	50.6%	Volume peak — competition begins to intensify
Sep–Nov 2023	1,257	41.0%	Dips — over-subscribed cycle
Jan 2024	1,509	27.0%	PEAK volume, LOWEST success — pipeline anomaly
Feb–Mar 2024	1,130	30.2%	Continued decline — high-competition environment



ANOMALY ALERT: January 2024 — 1,509 applications (all-time peak) with only 27.0% success (lowest recorded). A single high-demand Internship flooded the pipeline. Flag this cohort as anomaly in Week 3 modeling.

4.2 Peak Engagement Traffic by Sign-Up Period

Sign-up time-of-day analysis reveals learner engagement patterns across four daily periods:

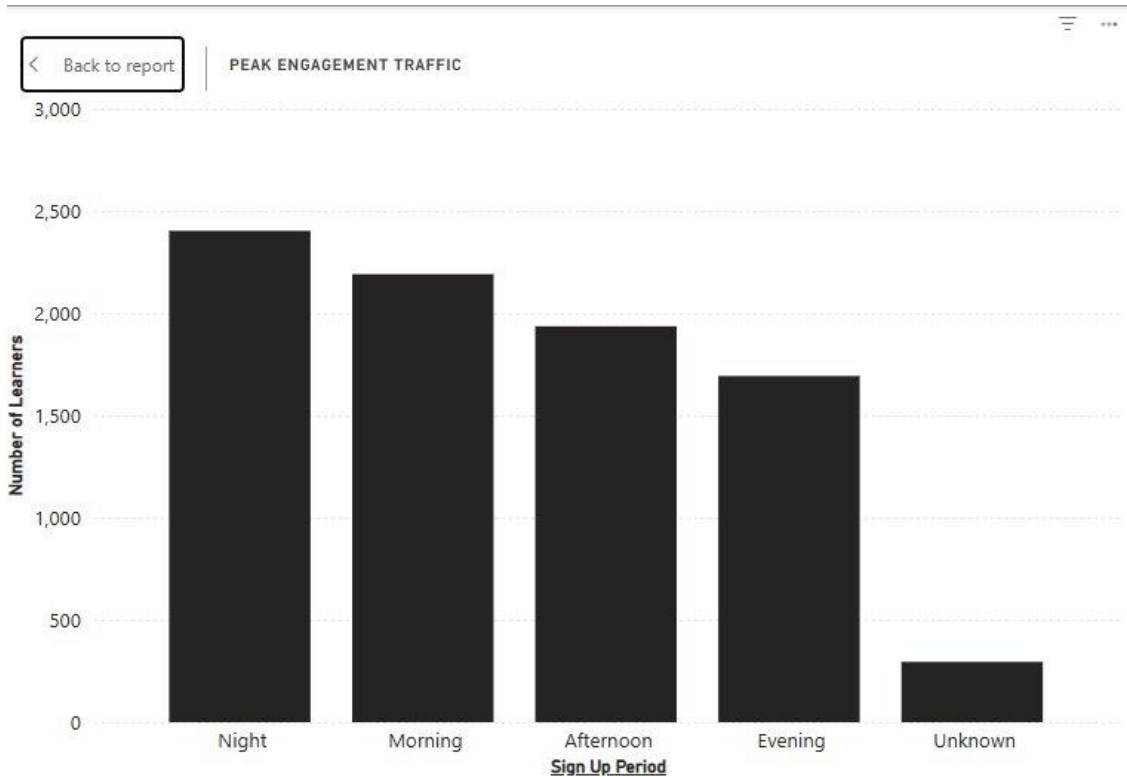


Figure 2: Peak Engagement Traffic by Sign-Up Period

Caption: Night (~2,380) and Morning (~2,200) are the dominant sign-up periods, followed by Afternoon (~1,950) and Evening (~1,700). This counter-intuitive night-peak suggests that a significant proportion of learners are based in non-US time zones where 'Night (US)' maps to local business hours — consistent with the 53.5% non-US learner population. The 'Unknown' category (~300 records) represents missing timestamp data flagged during cleaning.

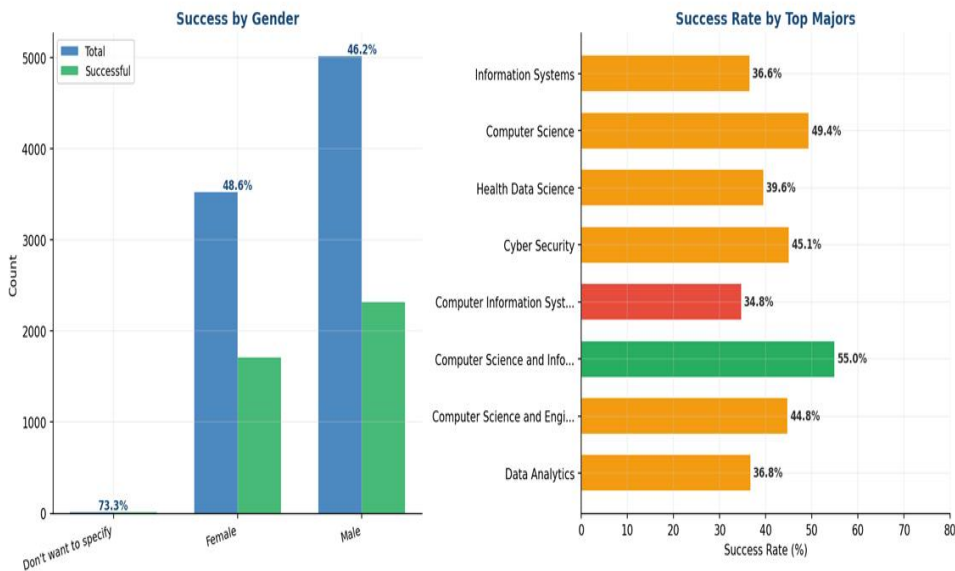
Sign-Up Period	Volume	Success Rate	Observation
Night (9 PM – 4 AM)	~2,380	~47%	Highest volume — reflects international time zones
Morning (5 AM – 11 AM)	~2,200	~47%	Second highest; early access pattern
Afternoon (12 PM – 4 PM)	~1,950	~47%	Professional hours — consistent performance
Evening (5 PM – 8 PM)	~1,700	~47%	After-work engagement peak
Unknown	~300	N/A	Missing timestamp — flagged in data cleaning

Section 5: Geographic & Demographic Analysis

5.1 Country-Level Success Rates

Country	Applications	% Total	Success Rate	Observation
United States	3,976	46.5%	39.5%	Below average — high competition
India	2,836	33.1%	49.2%	Slightly above platform average
Nigeria	760	8.9%	57.0%	Outperforms most large markets
Ghana	275	3.2%	61.5%	Strong success rate
Pakistan	219	2.6%	71.2%	Highest among high-volume markets
Egypt	50	0.6%	78.0%	Highest overall success rate
Kenya	38	0.4%	73.7%	Strong performer
Ethiopia	38	0.4%	68.4%	Above average
Bangladesh	65	0.8%	50.8%	Above average

Figure 7: Demographic & Academic Analysis



KEY INSIGHT: US students (46.5% of applications) achieve only 39.5% success — below the platform average. African nations (Egypt 78%, Kenya 73.7%, Nigeria 57%) and Pakistan (71.2%) show markedly higher success, suggesting geographic features capture systemic opportunity-access differences.

5.2 Application Delay Patterns by Status and Gender

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APPLICATION DELAY PATTERNS BY STATUS AND GENDER

status_description	Don'T Want To Specify	Female	Male	Other
Applied		98	74	
Dropped Out		48	46	
Rejected	0	50	59	0
Rewards Award		38	71	
Started	0	6	5	
Team Allocated	27	56	67	
Waitlisted	2	33	62	
Withdraw	98	70	23	

Figure 3: Application Delay Patterns by Status and Gender
Caption: The matrix shows that Team Allocated outcomes are dominated by Male (56) and 'Don't Want to Specify' (27) respondents in this filtered view. Withdrawn applicants skew Male (70) vs Female (98), suggesting different engagement dropout patterns by gender. Started status shows very low absolute numbers (6F, 5M), reflecting the small proportion of actively-engaged placements. Rejected and Dropped Out rows confirm high Internship-driven attrition across all genders.

5.3 Gender Distribution

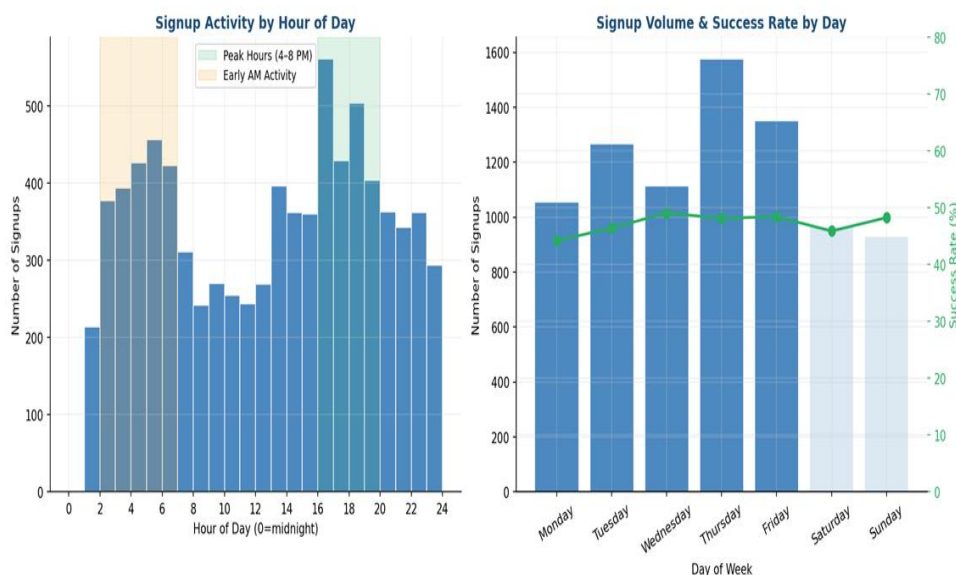
Gender	Total	Successful	Success Rate	Observation
Male	5,018	2,319	46.2%	Slightly below Female
Female	3,522	1,712	48.6%	Marginally higher success
Don't Want to Specify	15	11	73.3%	Small sample — not statistically significant

Section 6: Top Opportunities & Academic Analysis

6.1 Top 10 Most Applied Opportunities

Opportunity	Category	Apps	Successful	Rate	Flag
Career Essentials	Course	1,423	1,336	93.9%	Best overall
Data Visualization	Internship	980	125	12.8%	Bottleneck
Project Management	Internship	836	107	12.8%	Bottleneck
Healthcare Management	Internship	784	430	54.8%	OUTLIER
Digital Marketing	Internship	559	64	11.4%	Bottleneck
Data Viz Associate	Internship	551	121	22.0%	Avg
Business Consulting	Internship	483	78	16.1%	Below avg
Project Mgmt Assoc	Internship	478	107	22.4%	Avg
Innovation & Entrepreneurship	Internship	419	60	14.3%	Below avg
CPR/AED Certification	Course	389	328	84.3%	Strong

Figure 5: Signup Timing Analysis



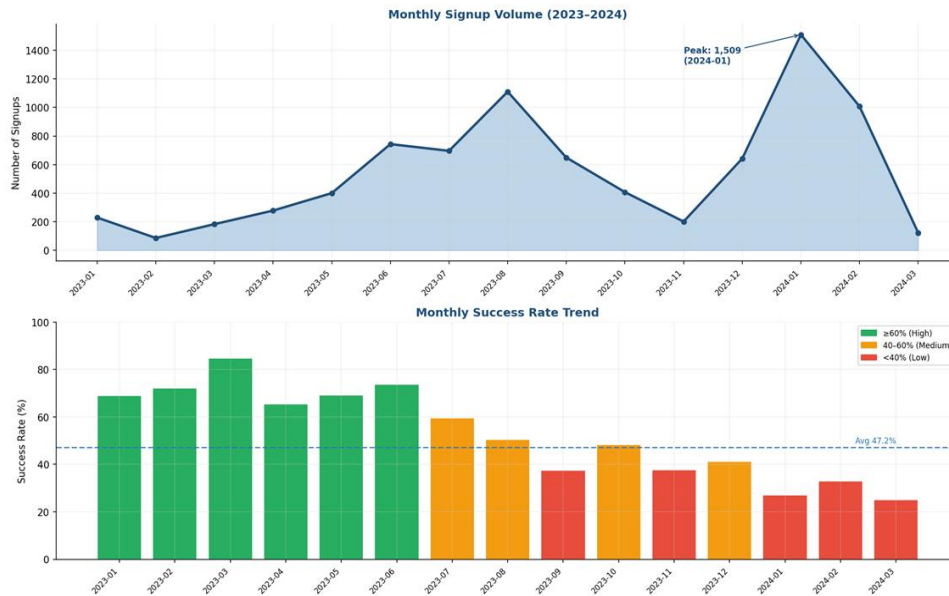
ANOMALY: Healthcare Management Internship shows 54.8% success — more than double the 21.5% Internship average. This outlier warrants separate treatment in predictive modeling.

6.2 Academic Major Performance

Academic Major	Records	Successful	Success Rate	Tier
Public Health	140	89	63.6%	HIGH
CS & Information Systems	211	116	55.0%	HIGH
Computer Science	1,052	520	49.4%	MEDIUM

Cyber Security	326	147	45.1%	MEDIUM	
Health Data Science	371	147	39.6%	MEDIUM	
Information Systems	2,158	789	36.6%	LOW	
Data Analytics	204	75	36.8%	LOW	
Computer Info Systems	250	87	34.8%	LOW	

Figure 3: Temporal Participation Trends



CONCERN: Information Systems — the most popular major with 2,158 records — has one of the lowest success rates (36.6%). This signals a potential mismatch between student interest and available placements.

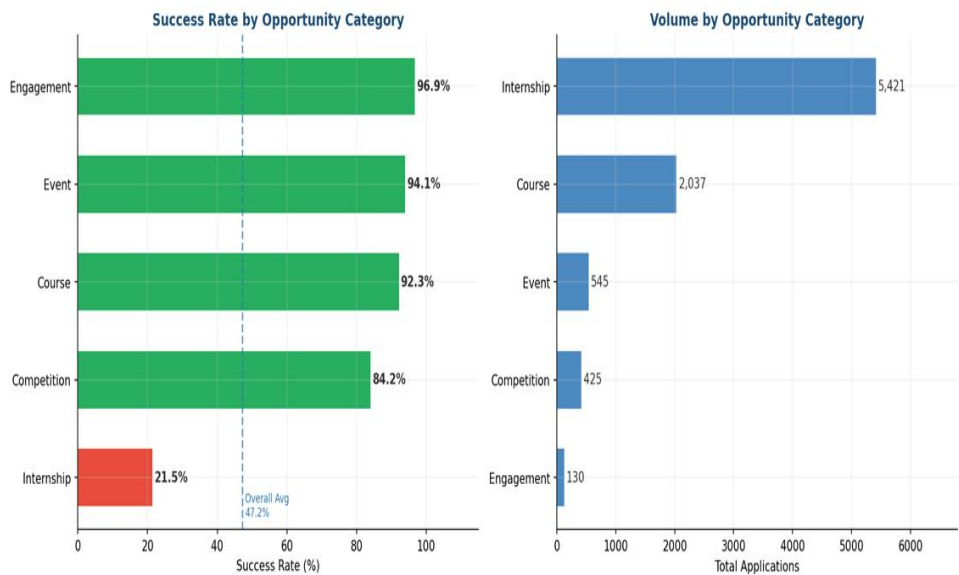
Section 7: Application Speed & Behavioral Patterns

7.1 The Deliberation Effect (Days_To_Apply)

Application timing reveals commitment and preparation patterns with measurable success impact:

Application Speed	Count	Success Rate	Interpretation
Same/Before signup	3,458	47.1%	Immediate — baseline performance
< 30 days after	1,709	46.9%	Quick — similar to baseline
30–90 days after	888	53.9%	BEST — deliberate timing, genuine fit
> 90 days after	2,058	45.4%	Late — slight underperformance

Figure 2: Opportunity Category Performance



EARLY SIGNAL: Students applying 30–90 days post-signup achieve the **HIGHEST** success (53.9%), outperforming both immediate (47.1%) and delayed (45.4%) applicants. Deliberate, well-timed applications reflect genuine fit and preparation.

Section 8: Interpretation of Findings

This section consolidates and merges findings from both the primary EDA and the supplementary interpretation analysis, identifying the strongest drivers of participation success, early warning signals, and consistent high-performers.

8.1 Strongest Drivers of Participation Success

#	Factor	Evidence	Implication for Modeling
1	Opportunity Category	21.5% (Internship) vs 96.9% (Engagement)	Strongest single predictor — encode as ordinal or one-hot; consider separate Internship model
2	Country of Origin	39.5% (US) vs 78.0% (Egypt)	Geographic feature captures systemic opportunity-access differences; include as encoded feature
3	Specific Opportunity	Healthcare Mgmt 54.8% vs Digital Marketing 11.4%	Opportunity-level fixed effects are important; Opportunity Name or ID should be included
4	Academic Major	Public Health 63.6% vs Comp. Info. Systems 34.8%	Major encodes discipline alignment; group rare majors into STEM / Health / Business / Other
5	Application Timing	30–90 day cohort 53.9% vs same-day 47.1%	Days_To_Apply is a useful behavioural predictor; bin into speed categories
6	Cohort Month/Year	Jan–Mar 2023 (75%) vs Jan 2024 (27%)	Signup_Month + Signup_Year captures competitive cycles; include as time features
7	Gender	Female 48.6% vs Male 46.2%	Marginal difference — include but expect low feature importance

8.2 Early Warning Signals for Low Engagement

- Internship applicants in Jan–Feb 2024: Success rate below 30% — over-subscription. Flag as high-risk cohort.
- Digital Marketing & Data Visualization Internships: Success rates of 11–13% — structural bottlenecks. Program managers should expand capacity or redirect applicants.
- Dropped Out cohort (617 students): Concentrated entirely in Internships (11.3% dropout rate). Post-placement engagement support is lacking.
- US applicants to Internships: Only 39.5% overall success — the largest group underperforms the platform average. High volume + high competition = low yield.
- Information Systems major (2,158 records, 36.6% success): Most popular major has one of the lowest success rates — mismatch between student interest and available placements.

8.3 Which Opportunity Types Consistently Attract High Participation

Category	Volume	Success	Verdict
Course	High (2,037)	92.3%	Best overall — high demand AND high success
Event	Medium (545)	94.1%	Near-perfect success; opportunity to scale
Engagement	Low (130)	96.9%	Highest success rate; expand capacity
Competition	Low (425)	84.2%	Strong performer; under-utilised
Internship	Dominant (5,421)	21.5%	High demand but capacity-constrained — core

			challenge
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Section 9: Week 3 Modeling Readiness

9.1 Dataset Readiness Confirmation

Criterion	Status	Detail
Target variable defined	READY	Is_Successful (binary 0/1) — Team Allocated or Started = 1
Date columns parsed	READY	All 6 date columns converted from string to DateTime; corruption fixed
Missing values addressed	READY	Text fields filled; date gaps flagged and documented; no silent NaN
Categorical encoding	READY	Category_Encoded and Gender_Encoded available for ML input
Derived features created	READY	13 new features engineered and validated
Temporal features available	READY	Signup_Year, Signup_Month, Apply_Month enable time-series cohort analysis
Class balance check	CAUTION	47.2% success vs 52.8% non-success — mild imbalance; monitor F1 score
Internship-specific behaviour	CAUTION	Internships behave differently — consider separate model or interaction terms
Opportunity Start Date gaps	FLAGGED	44.3% missing — DO NOT use as direct feature; use Engagement_Duration instead

9.2 Recommended Features for Week 3 Model

Feature	Type	Expected Importance	Rationale
Opportunity Category	Categorical	VERY HIGH	Strongest predictor — drives 21%–97% success range
Category_Encoded	Integer	VERY HIGH	Numeric version of Opportunity Category for tree-based models
Country (encoded)	Categorical	HIGH	39.5%–78% success range across countries
Opportunity Name / Id	Categorical	HIGH	Opportunity-level variation is large (11%–94%)
Major (grouped)	Categorical	MEDIUM	34%–64% range; group rare majors to reduce cardinality
Signup_Month	Integer	MEDIUM	Seasonal cohort patterns confirmed in EDA
Signup_Year	Integer	MEDIUM	Year-over-year platform growth changes competitive dynamics
Days_To_Apply	Float	MEDIUM	30–90 day window shows +7pp uplift vs same-day applicants
Gender_Encoded	Integer	LOW	Only 2.4pp difference Male vs Female — include but low weight expected
Signup_IsWeekend	Binary	LOW	No meaningful difference found in EDA — include for completeness
Engagement_Duration_Days	Float	MEDIUM	Proxy for opportunity intensity — available for

			~40% of records
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9.3 Recommended Model Approaches

- Logistic Regression (baseline): Interpretable, fast. Establish baseline accuracy and feature coefficients.
- Random Forest Classifier: Handles categorical features and class imbalance well. Generate feature importance chart.
- Gradient Boosted Trees (XGBoost/LightGBM): Best expected performer given mixed data types and non-linear relationships.
- Internship-specific sub-model: Given structural difference in Internship success rates, consider training a separate model on Internship records only.
- Class imbalance handling: Use stratified train/test split and evaluate using F1-score, AUC-ROC, and precision-recall curves — NOT just accuracy.

FINAL CONFIRMATION: The RIT_Cleaned_Dataset_v2.csv is fully prepared for Week 3 predictive modeling. Recommended first model: Random Forest with Opportunity Category, Country, Opportunity Name, and Days_To_Apply as primary features.

Section 10: Actionable Strategic Recommendations

Based on the EDA findings, the following recommendations are prioritised by impact and feasibility:

10.1 Capacity & Program Design

#	Recommendation	Action & Rationale
1	Expand High-Success Categories	Increase seat capacity for Engagements (96.9%), Events (94.1%), and Courses (92.3%). These formats are both high-success and in demand — scaling them reduces platform-wide rejection rates.
2	Address Internship Bottlenecks	Digital Marketing (11.4%) and Data Visualization (12.8%) have structural bottlenecks. Either expand mentor/host capacity or redirect students to bridge opportunities (e.g., related Courses) to maintain engagement.
3	Replicate Healthcare Model	Healthcare Management Internship achieves 54.8% success — more than double the Internship average. Investigate what structural factors (e.g., host partnership depth, applicant screening, support) enable this and replicate across other Internship tracks.

10.2 Learner Engagement & Timing

#	Recommendation	Action & Rationale
4	Promote Deliberate Application Timing	Students applying 30–90 days post-signup achieve 53.9% success. Implement a 'readiness prompt' feature — remind learners at Day 30 post-signup to apply. Avoid pushing immediate applications which show no advantage.
5	Support Jan–Feb 2024 Cohort	This cohort shows <30% success due to Internship over-subscription. Proactively offer alternative opportunities (Courses, Competitions) to keep this cohort engaged and reduce churn.
6	Reduce Dropout Rate	617 students dropped out — 100% from Internships. Launch post-placement check-ins at 2-week and 4-week marks. Early intervention at the onboarding stage reduces the 11.3% Internship dropout rate.

10.3 Predictive & Data-Driven Interventions

#	Recommendation	Action & Rationale
7	Build Success-Prediction Score	Use Week 3 model outputs to generate a 'success probability score' for each applicant. Flag high-risk profiles (Information Systems major + Internship + Jan 2024 cohort) for proactive advisor intervention before rejection.
8	Address Major–Opportunity Mismatch	Information Systems (2,158 students, 36.6% success) is the most popular major yet a low performer. Create a 'recommended opportunities' feature that maps student major to highest-success opportunity types.
9	Geographic Personalisation	US students (46.5% of platform) achieve only 39.5% success. Investigate whether this is driven by Internship concentration. African and South Asian markets (71–78% success) may benefit from additional opportunity creation in their regions.

10.4 Suggested Additional Visualisations for Week 3

The following two charts are recommended to deepen analytical clarity:

Chart	Type & Variables	Why It Adds Value
Suggested Chart 1	Heatmap: Success Rate by Country × Opportunity Category (Country on Y-axis, Category on X-axis, colour = success rate %)	Reveals whether US underperformance is universal or concentrated in Internships. Identifies geographic-category combinations with highest potential for targeted expansion.
Suggested Chart 2	Line Chart: Monthly Success Rate Trend (2023–2024) split by Opportunity Category	Shows whether the Jan 2024 decline is category-wide or Internship-specific. Enables seasonal capacity planning and early detection of future pipeline stress.

Week Two Complete

DATASET STATUS: FULLY PREPARED FOR WEEK 3

RIT_Cleaned_Dataset_v2.csv | N = 8,518 | 13 Engineered Features | Target Variable Defined
Recommended First Model: Random Forest with Opportunity Category, Country, Opportunity Name & Days_To_Apply